First and last name ________________________________ University ID: ________________

Connected and Autonomous Vehicles
Final test – June 22nd, 2026

PART 1 - For each question, there is only 1 correct answer

1. An autonomous driving car
   a) Can be classified in the 5+1 SAE levels, from L0 (no automation) to L5 (full automation)
   b) Can be designed only by computer scientists
   c) Cannot be more than Level 3
   d) Works by magic

2. Technology for connected vehicles
   a) Don't inherently expose security issues
   b) Don't inherently expose privacy issues
   c) Don't inherently expose safety issues
   d) Exposes itself to threats to security, privacy and safety

3. The adoption of simulators for the development of autonomous systems
   a) Is useless, because we can directly test real cars
   b) Is a necessary step to test AD software on realistic scenarios
   c) Is not possible because none of the existing simulator infrastructures is powerful enough
   d) Requires at least skills of economics

4. The *Perception* stage of an AD stack
   a) Is typically the one that processes most data
   b) Is the one that embeds Model Predictive Controllers
   c) Is the one that leverages low-level actuators such as PIDs
   d) Is the one that decides where the car should go

5. The *Planning* stage of an AD stack
   a) Is typically the one that processes most data
   b) Is the one that embeds Model Predictive Controllers
   c) Is the one that leverages low-level actuators such as PIDs
   d) Is the one that decides where the car should go

6. The *Control* stage of an AD stack
   a) Is typically the one that processes most data
   b) Is the one that embeds Model Predictive Controllers
   c) Is the one that leverages low-level actuators such as PIDs
   d) Is the one that decides where the car should go

7. The *Actuation* stage of an AD stack
   a) Is typically the one that processes most data
   b) Is the one that embeds Model Predictive Controllers
   c) Is the one that leverages low-level actuators such as PIDs
   d) Is the one that decides where the car should go

8. In the domain of autonomous systems, data-driven techniques such as Machine Learning
   a) Are not applicable, as we simply don't have enough data
   b) Are applicable as soon as we can provide data
   c) Are not applicable to camera frames
   d) Are not applicable to decision systems

9. One unsolved, open problem of autonomous systems:
   a) Is the lack of capability of detecting objects within camera frames
   b) Is the lack of capability of detecting objects within LiDAR frames
   c) Is the lack of capability of creating industrial processes to certify them
   d) Is the fact that vehicles cannot be connected to internet

10. Connected cities:
   a) Can provide AD engineers with "additional eyes" for the vehicles, but not with datasets
   b) Can provide AD engineers with datasets, but not with "additional eyes" for the vehicles
   c) Can provide AD engineers with datasets and "additional eyes" for the vehicles
   d) Cannot provide AD engineers with datasets, nor with "additional eyes" for the vehicles

First and last name ___________________________________ University ID: __________________

PART 2 – Open questions

11. List one possible overlap between the technologies that you saw in the course and your PhD research topic(s)

First and last name ___________________________________ University ID: __________________

12. [*Optional, confidential and not part of the evaluation]* Kindly provide a constructure feedback on the course so that I can improve it for the next years